

Rubric for Assignment 2: Machine Learning Model Development (20%)

This rubric is designed to assess students' performance in developing, evaluating, and comparing machine learning models as part of Assignment 2. Each criterion is evaluated on a 5-point Likert scale ranging from Poor (1) to Excellent (5).

Criteria	Excellent (5)	Good (4)	Satisfactory (3)	Needs Improvement (2)	Poor (1)	Weight	Calculation	Total
Model Selection	Selected multiple appropriate algorithms with strong justification based on data characteristics and problem type; demonstrates deep understanding of model suitability.	Selected suitable models with reasonable justification; shows understanding of model-data relationship.	Selected models that are mostly suitable but with limited justification.	Selected inappropriate or poorly justified models; minimal understanding of algorithm relevance.	Models chosen arbitrarily with no justification or understanding of data-model fit.	20%	/5 *20 =	
Model Development	Model fully developed, well-structured, and optimized; demonstrates excellent preprocessing, feature	Model developed correctly with minor issues; includes adequate preprocessing and tuning.	Model functional but lacks depth in feature engineering or hyperparameter tuning.	Model partially implemented or with errors; insufficient preprocessing steps.	Model incomplete or fails to run.	25%	/5 *25 =	

	engineering, and tuning.							
Model Evaluation	Comprehensive evaluation using multiple metrics (accuracy, precision, recall, F1, ROC-AUC, etc.); results well-interpreted with insightful discussion.	Evaluated using relevant metrics and interpretations; results clear and correct.	Basic evaluation using limited metrics; interpretation somewhat generic.	Evaluation metrics applied incorrectly or with minimal analysis.	No meaningful evaluation or metrics presented.	20%	/5 *20 =	
Model Comparison	Excellent comparison of models with visualizations (confusion matrix, ROC curve, learning curve); strong reasoning for final model selection.	Good comparison using key metrics and graphs; justification mostly sound.	Limited comparison with few metrics or visuals; justification not clearly explained.	Minimal comparison; lacks visuals or reasoning.	No comparison performed or entirely incorrect interpretation.	20%	/5 *20 =	
Coding and Testing	Code well-organized, clean, documented, and reproducible; models tested thoroughly	Code functional and readable; minor inconsistencies in documentation or testing.	Code runs with minimal errors; limited testing or poor structure.	Code incomplete, poorly documented, or frequently fails.	No working code or unable to reproduce results.	15%	/5 *15 =	

	with consistent results.							
Total/100								
Total/100 *20								